

Communication protocol

S7 <-> EasyControl

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Version 1.5

22.03.2023

1. PC->S7 : Analog general values

Description	S7-DB	Offset	Bytes	Typ	Unit
Massflow - Feeder1 [kg/h]	6010	0	4	REAL	[kg/h]
	...				
Massflow - Feeder9	6010	32	4	REAL	[kg/h]
Massflow - Feeder10	6010	36	4	REAL	[kg/h]

Description	S7-DB	Offset	Bytes	Typ	Unit
Net weight - Feeder1	6010	40	4	REAL	[kg]
	...				
Net weight - Feeder9	6010	72	4	REAL	[kg]
Net weight - Feeder10	6010	76	4	REAL	[kg]

Description	S7-DB	S7-DBO	Bytes	Typ	Unit
Totalisator - Feeder 1	6010	80	4	REAL	[kg]
	...				
Totalisator - Feeder 9	6010	112	4	REAL	[kg]
Totalisator - Feeder 10	6010	116	4	REAL	[kg]

2. PC->S7 : Analog special values

Description	S7-DB	Offset	Bytes	Typ
Rotational speed- Feeder1	6011	0	4	REAL
	...			
Rotational speed- Feeder9	6011	32	4	REAL
Rotational speed- Feeder10	6011	36	4	REAL

Description	S7-DB	Offset	Bytes	Typ	
Paddlespeed - Feeder1	6011	40	4	REAL	[0..100%]
	...				
Paddlespeed - Feeder9	6011	72	4	REAL	[0..100%]
Paddlespeed - Feeder10	6011	76	4	REAL	[0..100%]

Description	S7-DB	S7-DBO	Bytes	Typ	
Doserperformance- 1	6011	80	4	REAL	[kg/h]
	...				
Doserperformance- 9	6011	112	4	REAL	[kg/h]
Doserperformance- 10	6011	116	4	REAL	[kg/h]

Description	S7-DB	Offset	Bytes	Typ
Reserved - Feeder1	6011	120	4	REAL
	...			
Reserved - Feeder9	6011	152	4	REAL
Reserved - Feeder10	6011	156	4	REAL

Description	S7-DB	Offset	Bytes	Typ
Refill alarm limit - Feeder1	6011	160	4	REAL
	...			
Refill alarm limit - Feeder9	6011	192	4	REAL
Refill alarm limit - Feeder10	6011	196	4	REAL

Description	S7-DB	Offset	Bytes	Typ
Drive command- Feeder1	6011	200	4	REAL
	...			
Drive command- Feeder9	6011	232	4	REAL
Drive command- Feeder10	6011	236	4	REAL

Description	S7-DB	Offset	Bytes	Typ	
Tara weight - Feeder1	6011	240	4	REAL	[kg]
	...				
Tara weight - Feeder9	6011	272	4	REAL	[kg]
Tara weight - Feeder10	6011	276	4	REAL	[kg]

Description	S7-DB	Offset	Bytes	Typ
Sampe time - Feeder1	6011	320	4	REAL
	...			
Sampe time - Feeder9	6011	352	4	REAL
Sampe time - Feeder10	6011	356	4	REAL

Description	S7-DB	Offset	Bytes	Typ
Prop. Gain - Feeder1	6011	360	4	REAL
	...			
Prop. Gain - Feeder9	6011	392	4	REAL
Prop. Gain - Feeder10	6012	396	4	REAL

Description	S7-DB	Offset	Bytes	Typ
Massflow filter - Feeder1	6011	400	4	REAL
	...			
Massflow filter - Feeder9	6011	432	4	REAL
Massflow filter - Feeder10	6011	436	4	REAL

Description	S7-DB	Offset	Bytes	Typ
Alarm noise limit - Feeder1	6011	440	4	REAL
	...			
Alarm noise limit - Feeder9	6011	472	4	REAL
Alarm noise limit - Feeder10	6011	476	4	REAL

3. PC->S7 : Alarmstatus

Alarme

Description	Bit Nr.	
Massflow +	0	Massflow too high
Massflow -	1	Massflow too low
Drive command +	2	Drive command too high
Drive command -	3	Drive command too low
reserved	4	
feeder empty	5	
Load cell overload	6	
reserved	7	
Refill alarm	8	
reserved	9	
reserved	10	
Encoder error	11	
reserved	12	
reserved	13	
reserved	14	
reserved	15	
reserved	16	
reserved	17	
Load cell error	18	
reserved	19	
Ext. Alarm	20	
Line stop	21	
Sync. Error	22	
Load cell noise	23	
reserved	24	
reserved	25	
reserved	26	
reserved	27	
reserved	28	
Dose performance +	29	
Dose performance -	30	
reserved	31	

Description	S7-DB	Offset	Bytes	Typ
Alarm - Feeder1	6012	0	4	DWORD
		...		
Alarm - Feeder9	6012	32	4	DWORD
Alarm - Feeder10	6012	36	4	DWORD

4. PC->S7 : Feeder Processstatus

Status

Description	Bit Nr.	
No alarm pending	0	
reserved	1	
feeder running	2	
gravimetric mode active	3	
volumetric mode active	4	
Empty feeder running	5	1 : Empty feeder is running
refill disabled	6	disabled -> feeder empty
refill active	7	
calibration running	8	
reserved	9	
reserved	10	
reserved	11	
reserved	12	
reserved	13	
reserved	14	
reserved	15	
reserved	16	
reserved	17	
local mode active	18	feeder is in local mode
line mode active	19	feeder is in line mode
reserved	20	
reserved	21	
reserved	22	
reserved	23	
reserved	24	
reserved	25	
reserved	26	
reserved	27	
reserved	28	
reserved	29	
reserved	30	
reserved	31	

Description	S7-DB	Offset	Bytes	Typ
Status - Feeder1	6012	40	4	DWORD
	...			
Status - Feeder9	6012	72	4	DWORD
Status - Feeder10	6012	76	4	DWORD

5. S7->PC : Analog general values

Description	S7-DB	Offset	Bytes	Typ	Unit
Setpoint - Feeder1	6020	0	4	REAL	[kg/h]
		...			
Setpoint - Feeder9	6020	32	4	REAL	[kg/h]
Setpoint - Feeder10	6020	36	4	REAL	[kg/h]

Description	S7-DB	Offset	Bytes	Typ	Unit
Paddlespeed - Feeder1	6020	40	4	REAL	[0..100%]
		...			
Paddlespeed - Feeder9	6020	72	4	REAL	[0..100%]
Paddlespeed - Feeder10	6020	76	4	REAL	[0..100%]

6. S7->PC : Analog special values

	S7-DB	Offset	Bytes	Typ
Dose performance - 1	6021	0	4	REAL
	...			
Dose performance - 9	6021	32	4	REAL
Dose performance - 10	6021	36	4	REAL

	S7-DB	Offset	Bytes	Typ	Unit
Refill limit min- 1	6021	40	4	REAL	[kg]
	...				
Refill limit min - 9	6021	72	4	REAL	[kg]
Refill limit min - 10	6021	76	4	REAL	[kg]

	S7-DB	Offset	Bytes	Typ	Unit
Refill limit max- 1	6021	80	4	REAL	[kg]
	...				
Refill limit max - 9	6021	112	4	REAL	[kg]
Refill limit max - 10	6021	116	4	REAL	[kg]

	S7-DB	Offset	Bytes	Typ
Massflow alarm high- 1	6021	120	4	REAL
	...			
Massflow alarm high - 9	6021	152	4	REAL
Massflow alarm high- 10	6021	156	4	REAL

	S7-DB	Offset	Bytes	Typ
Massflow alarm low- 1	6021	160	4	REAL
	...			
Massflow alarm low - 9	6021	192	4	REAL
Massflow alarm low - 10	6021	196	4	REAL

	S7-DB	Offset	Bytes	Typ	Unit
Refill time - 1	6021	200	4	REAL	[s]
	...				
Refill time - 9	6021	232	4	REAL	[s]
Refill time - 10	6021	236	4	REAL	[s]
	S7-DB	Offset	Bytes	Typ	Unit
Refill alarm limit - 1	6021	240	4	REAL	[kg]
	...				
Refill alarm limit - 9	6021	272	4	REAL	[kg]
Refill alarm limit - 10	6021	276	4	REAL	[kg]

	S7-DB	Offset	Bytes	Typ	Unit
Alarm delay - 1	6021	280	4	REAL	[s]
	...				
Alarm delay - 9	6021	312	4	REAL	[s]
Alarm delay - 10	6021	316	4	REAL	[s]

	S7-DB	Offset	Bytes	Typ	Unit
Start delay - 1	6021	320	4	REAL	[s]
	...				
Start delay - 9	6021	352	4	REAL	[s]
Start delay - 10	6021	356	4	REAL	[s]

	S7-DB	Offset	Bytes	Typ	Unit
Sample time - 1	6021	360	4	REAL	[s]
	...				
Sample time - 9	6021	392	4	REAL	[s]
Sample time - 10	6021	396	4	REAL	[s]

	S7-DB	Offset	Bytes	Typ
Prop. gain - 1	6021	400	4	REAL
	...			
Prop. gain - 9	6021	432	4	REAL
Prop. gain - 10	6021	436	4	REAL

	S7-DB	Offset	Bytes	Typ
Massflow filter - 1	6021	440	4	REAL
	...			
Massflow filter - 9	6021	472	4	REAL
Massflow filter - 10	6021	476	4	REAL

	S7-DB	Offset	Bytes	Typ
Alarm noise limit - 1	6021	480	4	REAL
	...			
Alarm noise limit - 9	6021	512	4	REAL
Alarm noise limit - 10	6021	516	4	REAL

7. S7->PC : Command control

Control

Description	Bit Nr.	
Start feeder	0	start feeder
Stop feeder	1	stop feeder
reserved	2	
reserved	3	
reserved	4	alarm clear
Alarm clear	5	alarm clear
gravimetric	6	set feeder to gravimetric mode
volumetric	7	set feeder to volumetric mode
Feeder empty start	8	Start empty feeder
Refill disable	9	Refill disable / feeder empty
Refill start	10	Start refill / Only if Refill enable is set
Refill stop	11	Stop refill / if refill is active
Tare	12	tare feeder
Calibrate	13	calibrate feeder
Totalizer reset	14	reset totalizer
local mode	15	set feeder to local mode
line mode	16	set feeder to line mode
reserved	17	
reserved	18	
reserved	19	
reserved	20	
reserved	21	
reserved	22	
reserved	23	
reserved	24	
reserved	25	
reserved	26	
reserved	27	
reserved	28	
reserved	29	
reserved	30	
reserved	31	

Description	S7-DB	Offset	Bytes	Typ
Control - Feeder1	6022	0	4	DWORD
	...			
Control - Feeder9	6022	32	4	DWORD
Control - Feeder10	6022	36	4	DWORD

8. S7->PC : Line command control

Description	Bit Nr.		
start line	0		
stop line	1		
Totalizer reset	2		
reserved	3		
reserved	4		
reserved	5		
reserved	6		
reserved	7		
reserved	8		
reserved	9		
reserved	10		
set alarm 1	11	set alarm 1	flashlight on
reserved	12		
reserved	13		
reserved	14		
clr alarm 1	15	clr alarm 1	flashlight off
reserved	16		
reserved	17		
reserved	18		
reserved	19		
reserved	20		
reserved	21		
reserved	22		
reserved	23		
reserved	24		
reserved	25		
reserved	26		
reserved	27		
reserved	28		
reserved	29		
reserved	30		
reserved	31		

Description	S7-DB	Offset	Bytes
Line control	6022	40	4

9. PC->S7 : General communication parameters

Description	S7-DB	Offset	Bytes	Typ
Communication counter	6030	0	4	DWORD

Counter will be incremented by PC in order to assure communication is still alive

10.PC->S7 : Line process status

Description	Bit Nr.
reserved	0
reserved	1
reserved	2
reserved	3
reserved	4
reserved	5
reserved	6
reserved	7
reserved	8
reserved	9
reserved	10
reserved	11
reserved	12
reserved	13
reserved	14
reserved	15
reserved	16
reserved	17
reserved	18
reserved	19
reserved	20
reserved	21
reserved	22
reserved	23
reserved	24
reserved	25
reserved	26
reserved	27
reserved	28
reserved	29
reserved	30
reserved	31

Description	S7-DB	Offset	Bytes	Typ
Line process status	6030	4	4	DWORD

11. PC->S7 : Line alarm status

Description	Bit Nr.	
reserved	0	
reserved	1	
reserved	2	
reserved	3	
reserved	4	
alarm1 activated	5	flash light on ?
reserved	6	
reserved	7	
reserved	8	
reserved	9	
reserved	10	
reserved	11	
reserved	12	
reserved	13	
reserved	14	
reserved	15	
reserved	16	
reserved	17	
reserved	18	
reserved	19	
reserved	20	
reserved	21	
reserved	22	
reserved	23	
reserved	24	
reserved	25	
reserved	26	
reserved	27	
reserved	28	
reserved	29	
reserved	30	
reserved	31	

Description	S7-DB	Offset	Bytes	Typ
Line alarm status	6030	8	4	DWORD

12. Conversion from g/10s to kg/h

$$x \cdot \frac{g}{10s} = \frac{x}{10} \frac{[g]}{[s]} = \frac{x}{10} \cdot \frac{1000}{1000} \cdot \frac{3600}{3600} \frac{[g]}{[s]} = \frac{x}{10} \cdot \frac{3600}{1000} \cdot \frac{1000}{3600} \frac{[g]}{[s]} = \frac{x}{10} \cdot \frac{3600}{1000} \frac{[kg]}{[h]} =$$

$$\frac{36}{100} \cdot x \frac{[kg]}{[h]} = 0,36 \cdot x \frac{[kg]}{[h]}$$

13. Conversion from kg/h to g/10s

$$x \frac{[kg]}{[h]} = \frac{[100]}{[36]} \cdot x \frac{[g]}{[10s]}$$